Year 8
Mathematics

Unit 3: Working with Powers

3.2 More Simplifying

More Simplifying

STARTER · 5 MIN

Date: _____ **Title: More Simplifying**

LEARNING OBJECTIVES

- To be able to simplify expressions and use index laws in calculations

STARTER *Answer in your book. No calculators.*

1. Work out 3^5

2. Work out 2^6

3. Simplify $x \times x$

4. Simplify $4a \times 5b$

Starter — Answers

STARTER

Tick or correct in green pen

1. **243**
2. **64**
3. x^2
4. **20*ab***

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can collect like terms correctly	☐
SILVER	I can simplify expressions with powers	☐
GOLD	I can understand the meaning of an identity	☐

Tick each box in the plenary when you can do it on your own.

Introduction — More Simplifying

INTRODUCTION · 3 MIN

Index laws work for letters too: multiply the numbers, then use the laws.

Multiply

Add the powers.

$x^2 \times x^3 = x^5$

Divide

Subtract the powers.

$x^6 \div x^2 = x^4$

Power of a power

Multiply the powers.

$(x^2)^3 = x^6$

Identity ≡

True for every value.

$2x + 3x \equiv 5x$

Examples 1 and 2 — Multiplying Terms with Powers

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint Multiply the numbers Add the powers

Example 1

Simplify

$6x^2 \times 4x^3$

Example 2

Simplify

$3x^2 \times 4x^5$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Numbers × numbers, add the powers

$6 \times 4 = 24$

$x^2 \times x^3 = x^5$

Answer $= 24x^5$

Tip: never add the 6 and 4.

Example 2

Numbers × numbers, add the powers

$3 \times 4 = 12$

$x^2 \times x^5 = x^7$

Answer $= 12x^7$

Tip: 2 + 5 = 7.

Task 1

PRACTICE · 9 MIN

BRONZE *Simplify*

Hint Multiply the numbers Add the powers

1. $x^3 \times x^4$

2. $2a \times 5a^2$

3. $3x^4 \times 8x$

4. $4y^2 \times 2y^3$

5. $5m^3 \times m^2$

★ $2p^2 \times 3p^2 \times p$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 x^7

2 $10a^3$

3 $24x^5$

4 $8y^5$

5 $5m^5$

★ $6p^5$

Examples 3 and 4 — Dividing and Powers of Brackets

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Divide: subtract powers $(3x^2)^2$: square the 3 too

Example 3

Simplify

$$\frac{18x^5}{6x^2}$$

Example 4

Simplify

$(3x^2)^2$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Divide the numbers, subtract the powers

$18 \div 6 = 3$

$x^5 \div x^2 = x^3$

Answer $= 3x^3$

Tip: 5 − 2 = 3.

Example 4

Square everything in the bracket

$(3x^2)^2 = 3x^2 \times 3x^2$

$\mathbf{= 9x^4}$

Tip: $3^2 = 9$, not 6.

Task 2

PRACTICE · 9 MIN

SILVER *Simplify*

Hint Divide: subtract powers $(3x^2)^2$: square the 3 too

1. $48x^6 \div 2x$

2. $48x^8 \div 2x^3$

3. $\dfrac{20a^7}{4a^3}$

4. $(2x^3)^2$

5. $(5y)^2$

★ $(x^2)^4$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $24x^5$

2 $24x^5$

3 $5a^4$

4 $4x^6$

5 $25y^2$

★ x^8

Examples 5 and 6 — Identities and Areas

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint ≡ : true for all values Expand, then collect

Example 5

(3 marks)

Show that

$3y(5y^2 + 4y) + 2y^2(1 + 3y)$

$\equiv 21y^3 + 14y^2$

Example 6

(2 marks)

A shape is made from two rectangles, each $x + a$ wide and a high.

Write its area in terms of x and a.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Expand both, then collect

$15y^3 + 12y^2 + 2y^2 + 6y^3$

$\mathbf{= 21y^3 + 14y^2}$

Tip: $y \times y^2 = y^3$.

Example 6

Width × total height

Height $= a + a = 2a$

Area $= (x + a) \times 2a$

$\mathbf{= 2ax + 2a^2}$

Tip: multiply every term.

Task 3

PRACTICE · 9 MIN

GOLD *Exam-style: simplify and show*

Hint ≡ : true for all values Expand, then collect

1. $x(x^2 + 3) + 2x^3$

2. $2a^2(a + 4) - a^3$

3. Is $4x \times 2x \equiv 8x^2$?

4. Is $(2x)^3 \equiv 6x^3$?

5. Area of a square of side $3x^2$

★ Simplify $(2x^2)^3 \div 4x$

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1 $3x^3 + 3x$

2 $a^3 + 8a^2$

3 Yes

4 No: $8x^3$

5 $9x^4$

★ $2x^5$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Simplify $6x^2 \times 4x^3$

SILVER Simplify $48x^6 \div 2x$

GOLD Simplify $(3x^2)^2$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $24x^5$

SILVER $24x^5$

GOLD $9x^4$

Exam Practice — Multiplying Powers

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint Multiply the numbers Add the powers

Question 1

(2 marks)

Simplify $3x^2 \times 4x^5$

.....

Question 2

(2 marks)

Simplify

(a) $a^4 \times a^3$ (1)

.....

(b) $5m \times 2m^2$ (1)

.....

Exam Practice — Dividing and Brackets

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Divide: subtract powers
$(3x^2)^2$: square the 3 too

Question 3

(2 marks)

Simplify

$$\frac{18x^5}{6x^2}$$

.....

Question 4

(2 marks)

Simplify $(3x^2)^2$

.....

Exam Practice — Identities

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint ≡ : true for all values Expand, then collect

Question 5

(3 marks)

Show that

$2b(b^2 + 3b) - b(b^2 + 8b) \equiv b^3 - 2b^2$

.....

Question 6

(2 marks)

Which of these are identities?

A $4x \times 3x \equiv 12x^2$

B $(x^3)^2 \equiv x^5$

C $x^6 \div x^2 \equiv x^4$

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

12 — M1

$12x^7$ — A1

Tip: numbers ×, powers +.

Question 2

(a) a^7 — B1

(b) $10m^3$ — B1

Tip: $m = m^1$.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$18 \div 6 = 3$ **M1**

$\mathbf{3x^3}$ **A1**

Tip: subtract the powers.

Question 4

$3^2 = 9$ **M1**

$\mathbf{9x^4}$ **A1**

Tip: square the 3 as well.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$2b^3 + 6b^2 - b^3 - 8b^2$ **M1 M1**

$= b^3 - 2b^2$ **A1**

Tip: watch the − before *b*(...).

Question 6

A and C **B2**

B1 for one correct, none wrong

Tip: $(x^3)^2 = x^6$.